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Low-resource extraction with knowledge-aware pairwise prototype learning

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A B S T R A C T

Knowledge Extraction (KE) aims at extracting structured information from raw texts, such as *relation extraction* and *event extraction*. One of the major issues for KE is the *low-resource* problem due to deficient samples. Previous work addresses the low-resource issue mostly via data-driven methods, such as transfer learning, while neglecting *correlation knowledge* among classes. For example, inherent correlation of entailment between the relation pair and causality between the event pair can also be utilized for low-resource KE. Consequently, we propose to leverage correlation knowledge via pairwise prototype learning on the hypersphere with a novel framework called *Knowledge-aware Hyperspherical Prototype Network (K-HPN)*. K-HPN is able to recognize inherent correlation among classes, where each class is represented as a prototype on the hypersphere. The experimental results demonstrate that K-HPN outperforms previous methods of KE, particularly with low-resource training data regimes.

1. Introduction

Knowledge Extraction (KE) is a task aimed at extracting structural information from unstructured texts, such as relation extraction (RE) [1] and event extraction (EE) [2], which have been utilized for a broad range of important applications [3–5]. For instance, in the sentence "*Jack is married to the Iraqi microbiologist known as Dr. Germ.*", RE task should identify the relationship of the given entity pair ⟨Jack, Dr. Germ⟩ as '*husband_of*', and EE task should identify the event type as '*Marry*' where the word 'married' triggers the event and (Jack, Dr. Germ) are participants in the event as husband and wife respectively.

Low-resource Extraction. Traditional approaches [6–9] to KE mostly follow a supervised learning paradigm, which typically relies upon large sets of labeled data and they are unfortunately not readily available in real-life applications. For example, as depicted in Fig. 1(a), in the widely-used RE dataset, New York Times (NYT), a large portion of relations have a relatively small number of labeled sentences, and nearly 70% of the relations are long-tail [10]. Analogously, as shown in Fig. 1(b), for widely-used EE dataset, ACE-2005 corpus, nearly 60% of event types have less than 100 instances [6] and about 25% event types have less than 20 instances [11]. In fact, the maldistribution of RE and EE datasets impedes effective model learning to a great degree [12], since KE performance would decrease severely for long-tail relations and events.

As most KE models assume sufficient training corpus which are indispensable when learning versatile vectors for relations and events [13,14], it is difficult for relations or events which have extremely limited instances to achieve satisfactory performance. Hence it is crucial for KE models to be capable of extracting knowledge with low-resource training instances. Previous approaches address low-resource KE via distantly supervised learning [15,16], transfer learning [12,17], or meta learning [11,18]. Although these approaches can partly resolve low-resource issues, they are mostly data-driven, so that ignore inherent correlation knowledge among relations and events which may reduce the dependence on data.

Knowledge-aware Extraction. Intuitively, inherent correlation knowledge in the class label space can be suitable guidance

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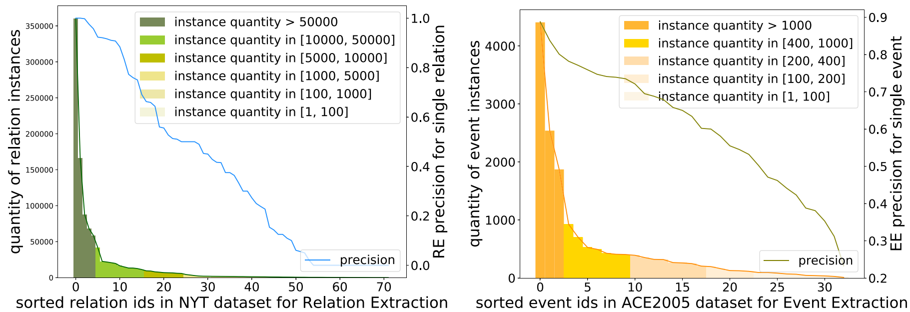


(a) Low-resource issues in RE. (b) Low-resource issues in EE.

Relation Type	Relation Instance
has cause (118 instances)	Colbert owned 25 *slaves* before the *American Civil War* broke out.
⇧ **entails**	
has immediate cause (7 instances)	The *Battle of Fort Sumter* is the first fighting of the *American Civil War.*

Event Type	Event Instance
Attack (461 instances)	A Protestant threw a gasoline *bomb* to a back yard.
⇩ **causes**	
Injure (53 instances)	An elderly man was *hospitalized* after a bomb landed in his back yard.

(c) Low-resource RE with correlation of relation entailment. (d) Low-resource EE with correlation of event causality.

Fig. 1. Illustration of utilizing correlation knowledge to benefit low-resource knowledge extraction tasks.

to boost low-resource performance [10,19]. For example, as seen in Fig. 1(c), the relation *has immediate cause* with only 7 instances `entails` *has cause* with 118 instances. The low-resource relation should obtain veracious embedding, if the model is able to detect the fact of ⟨*has immediate cause* $\xrightarrow{\text{entails}}$ *has cause*⟩, and correlation knowledge of relation entailment can be transferred from the high-resource relation (*i.e.*, *has cause*) to the low-resource one (*i.e.*, *has immediate cause*) [10]. Analogously, EE suffers identical difficulty, as depicted in Fig. 1(d), *Attack* (with 461 instances) has much more labeled samples than *Injure* (with 53 instances), and there veritably exists the fact of ⟨*Attack* $\xrightarrow{\text{causes}}$ *Injure*⟩. Correlation knowledge of event causality can be utilized to enhance extraction of low-resource events, even if the sample data are extremely limited [19].

Knowledge-aware Pairwise Prototype Learning for Low-resource Extraction. Following this motivation, we propose to resolve low-resource knowledge extraction problems in virtue of utilizing correlation knowledge of relation entailment and event causality. To this end, we propose a novel model called *Knowledge-aware Hyperspherical Prototype Network (K-HPN)* to (1) *learn robust representation for each class*, and (2) *utilize inherent correlation knowledge between the class pair*. In K-HPN, each class are embedded as a prototype which is defined as the average output feature vector over training samples per class, and it is proven to be robust in low-resource scenarios [20]. Different from vanilla Prototypical Networks (PNs) [20] where each class are embedded with part of instances in Euclidean space, we compute the prototype for each class with all of its instance embeddings with a maximum threshold. Furthermore, we also position prototypes on the hypersphere in hyperspherical output space, so as to utilize correlation knowledge among classes, inspired by Hyperspherical Prototype Networks (HPNs) [21]. HPNs demonstrate that *hyperspherical output spaces* have advantages in incorporating *correlation knowledge* in prototype construction, and they even propose that prototypes do not need to be inferred from data with relying on hyperspherical output spaces, which is naturally suitable for low-resource scenarios. Besides, [22] also demonstrate that hyperspherical output spaces are able to sufficiently cluster features of a class well, so as to obtain a robust hyperspherical prototype for each class.

Specifically, K-HPN divides HPN into a source and a target hemisphere, where each source–target prototype pair (represents class pair) are distributed on the corresponding hemisphere respectively, *e.g.*, *Attack* prototype locates at source hemisphere and *Injure* prototype locates at target hemisphere. Furthermore, the instance quantity of source and target classes often varies greatly, which means that the source–target class pair are respectively high/low resource, thus provides the possibility of source–target knowledge transfer. The high-resource class may help low-resource one to obtain a sound representation via learning between source–target prototype pair, thus it can remit low-resource issues to some extent. We hypothesize that *pairwise prototype learning on the hypersphere with correlation knowledge is able to benefit low-resource knowledge extraction by reducing the dependence on data*. In summary, the main contributions of our work are as follows:

- To the best of our knowledge, it is the first work to utilize correlation knowledge among classes in low-resource KE tasks. For the sake of resolving the new tasks, we establish two new datasets, called `Entail-RE` and `Causal-EE`.
- We represent each class as a prototype and leverage correlation knowledge between the class pair by pairwise prototype learning on the hypersphere, with a novel *Knowledge-aware Hyperspherical Prototype Network (K-HPN).*
- The experimental results on newly-proposed `Entail-RE` and `Causal-EE` datasets demonstrate that K-HPN significantly outperforms traditional methods of knowledge extraction, especially in low-resource scenarios.

The next section defines the problem and presents the details of K-HPN architecture. Section 3 introduces the experiments and evaluation results. Section 4 reviews related work on low-resource KE and correlation reasoning in KE. Section 5 makes a conclusion of the paper and discusses the future work.

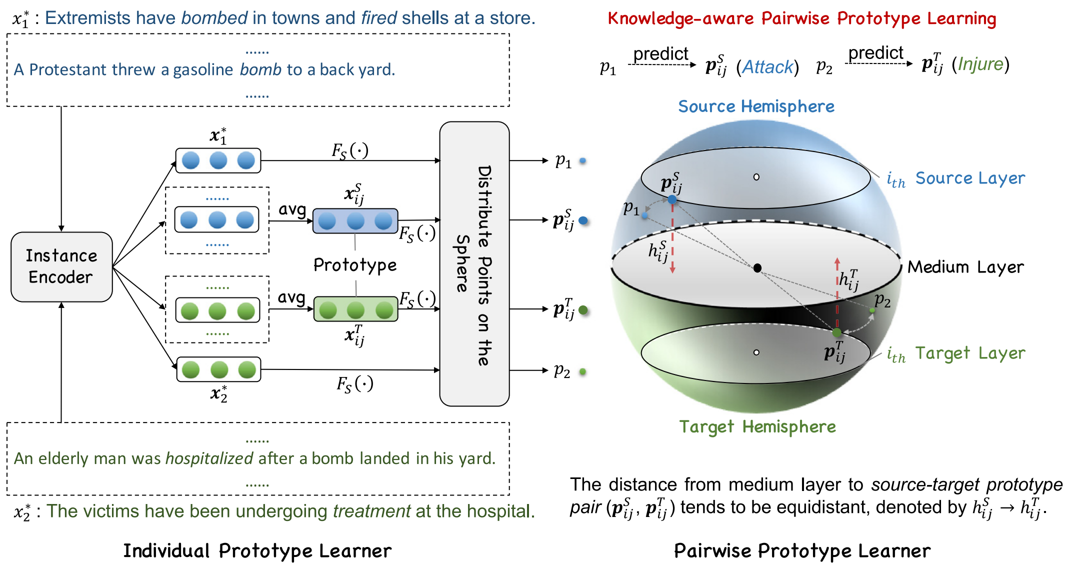


Fig. 2. Model Overview of K-HPN. x_k^* denotes the instance that input into Instance Encoder. $\boldsymbol{x}_k^*$ (*e.g.*, $k = 1, 2$), $\boldsymbol{x}_{ij}^S$ and $\boldsymbol{x}_{ij}^T$ respectively denote the instance embedding, source prototype and target prototype on plane space. $F_s(\cdot)$ is a function to distribute prototype points on a sphere. p_k (*e.g.*, $k = 1, 2$), $\boldsymbol{p}_{ij}^S$ and $\boldsymbol{p}_{ij}^T$ respectively denote the plane-space instance embedding, source prototype and target prototype on the hyperspherical space. h_{ij}^S and h_{ij}^T denote the distance of $\boldsymbol{p}_{ij}^S$ and $\boldsymbol{p}_{ij}^T$ to the medium layer respectively.

2. Methodology

2.1. Problem formalization

Knowledge Extraction. Supposing that we have a relation class set $\mathcal{R} = \{r_i | i \in [1, N]\}$, an event class set $\mathcal{E} = \{e_i | i \in [1, N]\}$, and corpus $\mathcal{T} = \{X_i | i \in [1, M]\}$ that contains M instances. Each instance X_i in $\mathcal{T}$ is denoted as a token sequence $X_i = \{x_i^j | j \in [1, L]\}$ with maximum L tokens. Our goal is to predict the relation and event label for each instance in corresponding corpus respectively.

Prototype Learning. The labeled class sets $\{\mathcal{R}, \mathcal{E}\}$ are represented as prototypes $\boldsymbol{P} = \{\boldsymbol{p}_i | i \in [1, N]\}$, and KE with correlation knowledge is implemented between the prototype pair. Generally, the goal of this work is to design an individual prototype learner $\boldsymbol{p} = F(X)$ that enables learning instance-level features and a pairwise prototype learner $\tilde{\boldsymbol{y}} = G(\boldsymbol{p})$ that enables utilizing class-level correlation knowledge.

2.2. Model overview

Different from traditional methods that regard KE tasks as classification [23–25] or sequence tagging [26,27] just on instance-level, we address KE tasks from a novel perspective. We revisit low-resource KE on class-level by representing each class as a prototype, and propose a *Knowledge-aware Hyperspherical Prototype Network (K-HPN)* for pairwise prototype learning with correlation knowledge. The model overview of K-HPN with running examples is shown in Fig. 2.

Generally, K-HPN can be divided into two modules, including (1) *a individual prototype learner to establish prototypes*, and (2) *a pairwise prototype learner to position source–target prototype pairs*. As seen from Fig. 2, for *individual prototype learner*, each class is represented by a prototype which is established via averaging its instance embeddings, and then we distribute these prototypes on the sphere. Further, for *pairwise prototype learner*, we partition prototypes on K-HPN into source–target pairs, and position prototype pairs on the source/target layer of corresponding hemisphere respectively, separated by a medium layer. For example, given a source–target prototype pair (*Attack*, *Die*) represented by $(\boldsymbol{p}_{ij}^S, \boldsymbol{p}_{ij}^T)$ which denotes the j_{th} source/target prototype located at the i_{th} source/target layer, $\boldsymbol{p}_{ij}^S$ and $\boldsymbol{p}_{ij}^T$ are expected to be equidistant from the medium layer. Note that for each source/target layer, there can be multiple prototypes on it. This allows K-HPN to implement knowledge-aware pairwise prototype learning on many-to-many prototype pairs, *e.g.*, *Die* event can be caused by more than one other events such as *Attack* as well as *Policy* and vice versa.

2.3. Individual prototype learner

The learning process of individual prototypes includes (1) *encoding instances* and (2) *establishing prototypes*.

Instance Encoder. We propose to encode instances by utilizing pre-trained word embedding and randomly initialized position embedding. Given a token sequence $X = \{x_1, \dots, x_L\}$, where x_i is the i_{th} token and L is the maximum length, the encoding includes two components: (1) word embedding $\boldsymbol{w} = \{\boldsymbol{w}_i \mid \boldsymbol{w}_i \in \mathbb{R}^{d_w}, i \in [1, L]\}$ to express semantic meanings, which are pre-trained via GloVe [28] with d_w–dimensional vector for each token, and (2) position embedding $\boldsymbol{v} \in \mathbb{R}^{3 \times d_p}$ to embed relative distances of labeled tokens in the sentence following [11,29]. For relation extraction, $\boldsymbol{v} = [\boldsymbol{v}_h, \boldsymbol{v}_t, \boldsymbol{v}_l]$ where the relative distances to the head and tail entity [29], as well as the token length are encoded with three d_p-dimensional vectors respectively. For event extraction, $\boldsymbol{v} = [\boldsymbol{v}_b, \boldsymbol{v}_e, \boldsymbol{v}_l]$ where distances from the trigger word to the beginning and ending of the sentence, as well as the sentence length are encoded with three d_p-dimensional vectors respectively [11].

We then achieve a final input embedding $\boldsymbol{x}$ for each token by concatenating its word embedding and position embedding, denoted by $\boldsymbol{x} = [\boldsymbol{w}, \boldsymbol{v}]$. The encoding of each instance is derived by a non-linear transformation $f(\cdot)$, which is denoted by

$$\boldsymbol{x}^* = f(\bar{\boldsymbol{x}}), \quad \bar{\boldsymbol{x}} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_L] \tag{1}$$

where we choose CNN [29] as $f(\cdot)$ to encode input embeddings $\bar{\boldsymbol{x}}$, and another sophisticated approaches can also be used, such as RNN [30] or BERT [31][1].

[1] We do not select BERT here because we concern that the pre-train model may import extra knowledge, which may weaken our knowledge-aware pairwise prototype learning.

Table 1
Statistics of our newly-proposed datasets `Entail-RE` with both relation type and entailment annotated and `Causal-EE` with both event type and causality annotated, compared with existing datasets. ET-Link: relation entailment link *between relation class pair*, and C-Link: event causality link *between event class pair*. Note that (1) T-REx and Wikidata have no annotation of relation entailment, and (2) although Causal-TB, MATRES, and EventCausality respectively have 318, 172, and 580 event causality links *between event instance pair*, they have no annotation of event types, let alone C-Link.

Datasets		#Doc	#Ins	#Class	#ET-Link/#C-Link
Relation	T-REx [32]	–	6.3*M*	642	*None*
	Wikidata [33]	–	741*k*	352	*None*
	Entail-RE	–	**220k**	**80**	**19**
Event	Causal-TB [34]	183	6.8*k*	*None*	–
	MATRES [35]	203	1.3*k*	*None*	–
	EventCausality [36]	25	0.8*k*	*None*	–
	ACE-2005[1]	589	16.4*k*	33	*None*
	TAC-KBP-2017[2]	167	4.4*k*	18	*None*
	FewEvent [11]	802	67.9*k*	100	*None*
	Causal-EE	–	**3.7k**	**80**	**112**

Prototype Establishment. We select prototypes to represent classes as it is proven to be robust in low-resource scenarios [20]. Different from vanilla prototype networks [20] where each prototype is obtained by calculating an average of *part of* instance embeddings in *Euclidean space*, we generate each prototype by averaging *all* instance embeddings in *hyperspherical space*. We distribute prototypes on the hypersphere as [21] demonstrate that hyperspherical prototypical networks are able to utilize *correlation knowledge* to construct prototypes. Besides, considering both accuracy and simplicity, we also set a threshold σ of maximum instance number, and randomly sample σ instances of the class with overfull instances.

Given a relation class r_i and an event class e_i with its instance encodings $\boldsymbol{x}^*_{r_i}$ and $\boldsymbol{x}^*_{e_i}$, the corresponding prototype embeddings of them are denoted by

$$\boldsymbol{p}^r_i = F_s\big(avg(\sum_{i=1}^{M_{r_i}} \boldsymbol{x}^*_{r_i})\big),\ \ \boldsymbol{p}^e_i = F_s\big(avg(\sum_{i=1}^{M_{e_i}} \boldsymbol{x}^*_{e_i})\big) \tag{2}$$

where $F_s(\cdot)$ is a function to distribute prototype points on a sphere [37], $avg(\cdot)$ is an average function, and M_{r_i} and M_{e_i} is the maximum instance number of relation r_i and event e_i respectively.

2.4. Pairwise prototype learner

The pairwise hyperspherical prototype learning is partitioned into two parts, including (1) *classification with hyperspherical prototypes* and (2) *extraction with pairwise prototype learning*.

Classifying Hyperspherical Prototypes. We propose to classify new instances by minimizing the angle between output vector $f(\tilde{\boldsymbol{x}}_i)$ and prototype $\boldsymbol{p}_{y_i}$ for the ground truth label y_i. Thus the classification loss $\mathcal{L}_c$ to minimize is denoted by

$$\begin{aligned}\mathcal{L}_c &= \sum_{i=1}^{N}\Big(1-\cos\theta_{f(\tilde{\boldsymbol{x}}_i),\boldsymbol{p}_{y_i}}\Big)^2\\ &= \sum_{i=1}^{N}\Bigg(1-\frac{\left|f(\tilde{\boldsymbol{x}}_i)\cdot\boldsymbol{p}_{y_i}\right|}{\left\|f(\tilde{\boldsymbol{x}}_i)\right\|\left\|\boldsymbol{p}_{y_i}\right\|}\Bigg)\end{aligned} \tag{3}$$

and the partial derivative of the loss function in Eq (3) with respect to $f(\tilde{\boldsymbol{x}}_i)$ is defined as

$$\begin{aligned}\frac{\partial\mathcal{L}_c}{\partial f(\tilde{\boldsymbol{x}}_i)} &= \frac{\partial\big(1-\cos\theta_{f(\tilde{\boldsymbol{x}}_i),\boldsymbol{p}_{y_i}}\big)^2}{\partial f(\tilde{\boldsymbol{x}}_i)}\\ &= 2\big(1-\cos\theta_{f(\tilde{\boldsymbol{x}}_i),\boldsymbol{p}_{y_i}}\big)\\ &\quad\Bigg(\frac{\cos\theta_{f(\tilde{\boldsymbol{x}}_i),\boldsymbol{p}_{y_i}}\cdot f(\tilde{\boldsymbol{x}}_i)}{\left\|f(\tilde{\boldsymbol{x}}_i)\right\|^2}-\frac{\boldsymbol{p}_{y_i}}{\left\|f(\tilde{\boldsymbol{x}}_i)\right\|\left\|\boldsymbol{p}_{y_i}\right\|}\Bigg)\end{aligned} \tag{4}$$

Our optimization aims for angular similarity between output encodings and class prototypes. Thus for a new data point $\tilde{\boldsymbol{x}}_i$, prediction is performed by computing the cosine similarity to all class prototypes and then select the class with the highest similarity, denoted by

$$\mathcal{C}^* = \underset{\mathcal{C}\in\{\mathcal{R},\mathcal{E}\}}{\operatorname{argmax}}(\cos\theta_{f(\tilde{x}_i),\boldsymbol{p}_{y_i}}) \tag{5}$$

Extraction with Pairwise Prototype Learning. By far, all prototypes are just required to be separated from each other randomly, as no correlation knowledge of classes are assumed. However, semantically-related classes should be relevant to each other in terms of prototype position. In order to infuse correlation knowledge into prototype position, we propose to utilize relation entailment and event causality between each prototype pair, which are expressed by source–target prototype pairs. We suppose that a source–target prototype pair should be located on opposite positions, meaning that they are located on the source and target layer respectively but with nearly equal absolute value of distance to the medium layer. To discover the source–target correlation among prototypes, we propose a score function to measure the likelihood that the i_{th} and j_{th} prototypes can compose a source–target prototype pair, denoted by

$$\mathcal{S}(\boldsymbol{p}_i,\boldsymbol{p}_j) = \frac{\lambda}{|D(\boldsymbol{p}_i)-D(\boldsymbol{p}_j)|} \tag{6}$$

where $D(\cdot)$ is a function to calculate distance from a prototype to the medium layer, $|\cdot|$ is an absolute function, and λ is a hyper-parameter. Apparently, larger $\mathcal{S}(\boldsymbol{p}_i,\boldsymbol{p}_j)$ means that the i_{th} and j_{th} prototypes are more likely to compose a source–target prototype pair. Note that if $(\boldsymbol{p}_i,\boldsymbol{p}_j)$ constitutes a source–target prototype pair, $\mathcal{S}(\boldsymbol{p}_i,\boldsymbol{p}_j)$ tends to 0, but not equal to 0, thus it would not be infinity. In fact, distance of a source–target prototype pair to the medium layer are usually unequal but similar during training.

Then, in order to learn correlation knowledge between the prototype pair, we introduce a loss function for pairwise learning between the prototype pair, defined as

$$\mathcal{L}_p = -\sum_{i=1}^{N}\sum_{j=1}^{N}\mathcal{S}(\boldsymbol{p}_i,\boldsymbol{p}_j),\ i\neq j \tag{7}$$

Our ultimate goal is to implement knowledge extraction with correlation knowledge, hence the final loss of K-HPN is integrated $\mathcal{L}_c$ for prototype classification with $\mathcal{L}_p$ for prototype-pair learning, denoted by

$$\mathcal{L} = \alpha\mathcal{L}_c + (1-\alpha)\mathcal{L}_p \tag{8}$$

where α is a coefficient between 0 and 1.

3. Experiments

The experiments seek to: (1) demonstrate that K-HPN with pairwise prototype learning can benefit low-resource KE; (2) assess the effectiveness of different strategies in K-HPN; and (3) provide evidence that utilizing correlation knowledge in low-resource KE is feasible. To this end, we explore the interplay between pairwise prototype learning and low-resource KE in two tasks: (1) *RE with relation entailment recognition*, and (2) *EE with event causality recognition*.

3.1. Dataset

To validate the effectiveness of K-HPN with pairwise prototype learning, we propose two new datasets specifically for KE tasks with correlation knowledge. The newly proposed datasets can be obtained from the following link: https://github.com/231sm/Reasoning_In_KE.

In order to evaluate the impacts of pairwise prototype learning on RE, we establish a new dataset `Entail-RE` with *relation entailment pairs*, such as ⟨*has immediate cause* $\xrightarrow{\texttt{entails}}$ *has cause*⟩, and ⟨*basin country* $\xrightarrow{\texttt{entails}}$ *country*⟩. We augment some RE datasets with relation entailment annotation, including *T-REx* [32] and *Wikidata* [33].

In order to evaluate the effects of pairwise prototype learning on EE, we establish a new dataset `Causal-EE` with *event causality pairs*, such as ⟨*Attack* $\xrightarrow{\texttt{causes}}$ *Injure*⟩, and ⟨*Collaboration* $\xrightarrow{\texttt{causes}}$ *Investment*⟩. We augment EE datasets with causality annotation, including *ACE-2005*[2], *TAC-KBP-2017*[3], and *FewEvent* [11]. We also extend event causality datasets with event type information, including *Causal-TB* [34], *MATRES* [35], and *EventCausality* [36].

The statistics of two newly-proposed datasets are shown in Table 1. Note that for each class pair, the instance quantity of two classes often varies greatly, which means that the two classes are respectively high/low resource. For example, in a relation entailment pair ⟨*has immediate cause* $\xrightarrow{\texttt{entails}}$ *has cause*⟩, relation *has cause* has 118 instances, while relation *has immediate cause* merely has 7 instances.

3.2. Baseline and setting

Baseline. In order to assess the performance of K-HPN, we compare K-HPN with baselines from the perspective of models and tasks respectively.

From the perspective of tasks. To demonstrate the influence of relation entailment recognition on RE, we compare K-HPN with traditional RE models without recognizing relation entailment, including CNN [29], PCNN [1], and KEGCN [10]. Simultaneously, in order to validate the influence of event causality recognition on EE, we compare K-HPN with some classic EE models without identification on event causality, including DMCNN [2], JRNN [38], and JMEE [39]. These KE models are based on CNN, RNN and GCN, which are proved effective to some extent, while ignore correlation knowledge among classes.

From the perspective of models. To verify the effectiveness of correlation and hyperspherical strategies in K-HPN, we compare K-HPN with vanilla Prototypical Network (PN) [20] and Hyperspherical Prototype Network (HPN) [21] respectively. Note that in vanilla PN, different from the paradigm of n-way-k-shot settings, which means that we construct prototypes by sampling only a fraction of instance embeddings with a fixed small number n for each of the k classes, we establish prototypes by averaging all instance embeddings of each class with a maximum threshold.

Table 2
Performance of low-resource relation extraction. Note that we also train on data subset with ratio of 25%, 50%, and 75% respectively.

Model	P	R	F
CNN [29]	0.7540	0.7109	0.7257
PCNN [1]	0.7898	0.7412	0.7645
KEGCN [10]	0.8074	0.7639	0.7911
K-HPN	**0.8524**	**0.8113**	**0.8386**
+25%	0.8082	0.7754	0.7989
+50%	0.8276	0.7967	0.8023
+75%	0.8398	0.8045	0.8156

Setting. With regard to settings of the training process, SGD [40] optimizer is used, with 30,000 iterations of training and 2,000 iterations of testing. The dimension of word embedding and position embedding are set to 50 and 30 respectively. In K-HPN, a dropout rate of 0.2 is used to avoid over-fitting, and the learning rate is 1×10^{-3}. The hyperparameters of σ, λ and α are set to 200, 1 and 0.5[4] respectively. We evaluate the performance of RE and EE with *Precision*(P), *Recall*(R) and $F1$ *Score*(F).

3.3. Evaluation on low-resource ke with correlation

This section is primarily intended to demonstrate that K-HPN with correlation knowledge can benefit low-resource KE. We compare K-HPN with baselines on two tasks of low-resource knowledge extraction, including (1) low-resource relation extraction and (2) low-resource event extraction.

Low-resource Relation Extraction. To verify that correlation knowledge can benefit low-resource RE, we compare K-HPN with some classic RE models, as shown in Table 2. A general inspection reveals that K-HPN outperforms traditional RE models based on both CNNs and GCNs. A possible explanation might be that K-HPN is more robust in low-resource RE, as it integrates all instances of each relation and is capable of learning entailment correlation between the relation pair. Notably, further inspection reveals that K-HPN can still achieve considerable improvement even with fewer instances as we randomly sample 25%, 50%, and 75% training data respectively. This result echos that K-HPN is more suitable for RE in low-resource scenarios, owing to its capability of pairwise learning between the prototype pair with *correlation knowledge* of relation entailment.

Low-resource Event Extraction. In order to validate that K-HPN with correlation knowledge can benefit low-resource EE, we compare K-HPN with traditional EE models, as shown in Table 3. Generally, we observe that K-HPN outperforms classic EE models based on CNNs, RNNs, and GCNs, demonstrating that the prototype-based model with causality correlation is capable of improving EE in low-resource scenarios. Analogously, we also train K-HPN with fewer data in low-resource EE tasks, and obtain similar comparison results to low-resource RE, indicating that K-HPN with pairwise prototype learning is superior to baseline models for low-resource EE. The possible reason for this is that K-HPN with pairwise prototype learning on event causality has less dependence on data by capturing event causality correlation.

Through analyzing the experimental results of low-resource RE and EE, we can conclude that *K-HPN with pairwise prototype learning is more competitive than traditional KE models*. The possible reason is that *traditional KE models are mostly data-driven, and greatly rely on data, while K-HPN is able to optimize data-independently with correlation knowledge. In low-resource scenarios, K-HPN can achieve better performance by reducing the dependence on data.*

[2] https://catalog.ldc.upenn.edu/LDC2006T06

[3] https://tac.nist.gov/2017/KBP/Event/index.html

[4] Note that the best performance is achieved when α is set to 0.5.

Table 3
Performance of low-resource event extraction. Note that we also train on data subset with ratio of 25%, 50%, and 75% respectively.

Model	P	R	F
DMCNN [2]	0.7033	0.6769	0.6914
JRNN [38]	0.7156	0.6831	0.7088
JMEE [39]	0.7491	0.7034	0.7418
K-HPN	**0.7889**	**0.7438**	**0.7732**
+25%	0.7421	0.7132	0.7399
+50%	0.7605	0.7204	0.7539
+75%	0.7713	0.7378	0.7694

Table 4
Comparison results of low-resource KE on prototype-based networks. (*cor* and *hyp* is the abbreviation of correlation and hyperspherical strategies respectively.).

Task	Model	P	R	F
RE	**K-HPN**	**0.8524**	**0.8113**	**0.8386**
	PN (w/o *hyp* & *cor*)	0.8019	0.7507	0.7764
	HPN (w/o *cor*)	0.8301	0.8085	0.8125
EE	**K-HPN**	**0.7889**	**0.7438**	**0.7732**
	PN (w/o *hyp* & *cor*)	0.7494	0.7178	0.7372
	HPN (w/o *cor*)	0.7675	0.7301	0.7619

3.4. Ablation study

This section is mainly intended to assess the effectiveness of different strategies in K-HPN, including learning with correlation knowledge and modeling on the hypersphere. To this end, we implement comparison experiments by respectively removing the correlation and hyperspherical strategies in K-HPN.

The Effectiveness of Correlation Strategies. To illustrate the effectiveness of correlation strategies, we compare K-HPN with two prototype-based models without correlation knowledge, including PN [20] and HPN [21]. Compared to these models, K-HPN considers inherent correlation of relation entailment and event causality, making it capable of learning correlation knowledge between the prototype pair.

Results for low-resource KE on prototype-based models are shown in Table 4. We observe that K-HPN outperforms vanilla PN and HPN in both RE and EE task, indicating that pairwise prototype learning with correlation knowledge facilitates low-resource problems in KE. The outperformance may be due to that *K-HPN is able to utilize inherent correlation between the prototype pair and transfer knowledge among them, especially from high-resource prototypes to low-resource ones, so that obtain more accurate prototypes*.

We then visualize prototypes learned from the three prototype-based models in Fig. 3. Seen from Fig. 3(c), K-HPN is able to precisely partition prototypes into source and target ones, showing that K-HPN can capture relation entailment and event causality well. For instance, *Injure*, *Die*, and *Arrest-Jail* are all caused by *Attack*, so the four prototypes are almost equidistant to the medium layer, while *Attack* prototype is opposite to the others. This is mainly because K-HPN considers correlation knowledge when training. Comparing Fig. 3(c) with 3(a) and 3(b), we can observe that PN and HPN without correlation knowledge randomly distinguish prototypes, as they hardly recognize inherent correlation between the prototype pair. Besides, they tend to shorten the distance among prototypes with correlation of relation entailment or event causality. For example in Figs. 3(a) and 3(b), *Attack*, *Injure* and *Die* are close to each other, making these semantically-related prototypes difficult to distinguish instead.

The Effectiveness of Hyperspherical Strategies. To demonstrate the effectiveness of hyperspherical strategies, we compare hyperspherical-modeling methods, *i.e.*, K-HPN and HPN, to PN with plane modeling. As seen in Fig. 3 and Table 4, both K-HPN and HPN show great advantages compared to PN. The outperformance may be owing to *the usage of hyperspheres as output spaces, and prototypes are defined with correlation knowledge via large margin separation, making it possible to position prototypes through data-independent optimization. Especially in low-resource scenarios, this way of optimization is able to greatly promote KE with knowledge-aware pairwise prototype learning, and reduce dependence on data.*

Table 5
Results of relation entailment recognition, where $\langle$*targetrelation* $\xrightarrow{\texttt{entails}}$ *sourcerelation*$\rangle$.

Source relation	*member of*	*producer*
Top 5 target relation candidates	member of sports team	composer
	member of political party	screenwriter
	cast member	librettist
	country of citizenship	contributor
	contributor	director of photography

3.5. Case study: Evaluation of utilizing correlation knowledge in low-resource KE

This section aims to provide evidence that utilizing correlation knowledge in low-resource KE is feasible, *i.e.*, *relation entailment* with RE and *event causality recognition* with EE.

Case 1: Relation Entailment Recognition. To demonstrate that K-HPN is able to identify relation entailment in RE, we select some source–target prototype pairs of relations, shown in Table 5.

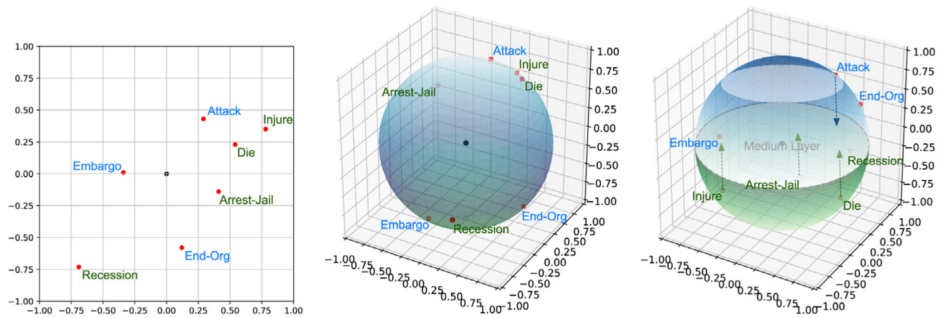
We list top 5 target relation candidates for two source relations, including *member of* and *producer*. For *member of*, the top 3 target relation candidates (*e.g.*, *member of sports team*) are exactly the ground truth, as they all entail *member of*. Furthermore, the other two relations are also semantically-related, and both of them are about the relationship between people and organization like *member of*. Similarly, for *producer*, the target relation candidates such as *composer*, *screenwriter* as well as *librettist* naturally entail *producer*, and the remaining two candidate relations are extremely relevant to *producer*. This reflects that K-HPN can commendably recognize entailment correlation among relations in low-resource RE.

Case 2: Event Causality Recognition. For the sake of proving that K-HPN is able to recognize event causality in EE, we illustrate causality correlation between the event pair. Visualization results for event causality recognition are shown in Fig. 4.

As seen from Fig. 4 based on the horizontal axis, we can observe that different source events have distinguished target event candidates. For example, *Sentence* is more likely to cause *Arrest-Jail* compared to other four candidate target events, and *Negotiation* is more likely to cause *Union*. Besides, seen from Fig. 4 based on the longitudinal axis, we can also observe that target events show differentiated attention to candidate source events. For instance, *Recession* is more likely caused by *Embargo* as well as *End-Org* (means *end organization*), and *Die* as well as *Injure* are both more likely caused by *Attack*. This illustrates that K-HPN is able to effectively capture causal correlation between the source–target event pair.

4. Related work

Low-resource Knowledge Extraction. Traditional approaches to low-resource KE mostly focus on enhancing model learning abilities and enhancing data utilization and extrapolation. Methods of enhancing learning abilities for low-resource KE mainly focus on *meta learning*. For example, [23] propose hybrid attention-based prototypical networks for noisy few-shot relation classification. [11] reformulate the event detection task with limited labeled data as a few-shot learning problem, and propose a



(a) PN. (b) HPN. (c) K-HPN.

Fig. 3. Visualization of prototypes. Note that we only show event prototypes here due to space limits.

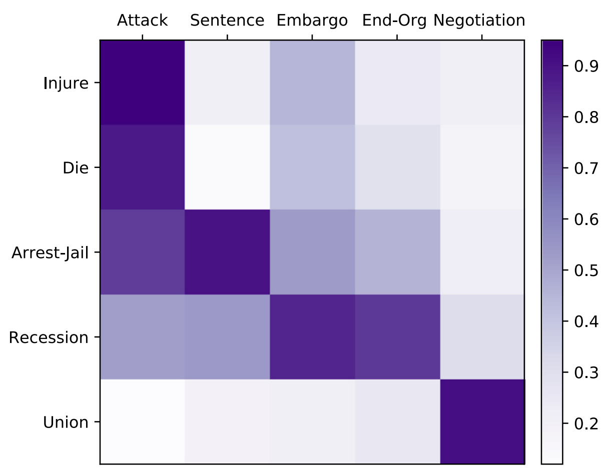


Fig. 4. Illustration of causality correlation between the event pair, where source and target events are located at the horizontal and longitudinal axis respectively with $\langle$*source event* $\xrightarrow{\text{causes}}$ *target event*$\rangle$. The darker the color, the greater the causality correlation.

dynamic-memory-based prototypical network to resolve it. These learn-to-learn approaches are mostly data-driven and short of utilizing correlation knowledge. Besides, some methods of enhancing data utilization and extrapolation for low-resource KE are based on *supervised or distantly supervised learning*. For instance, [15] introduce a RE approach based on distant supervision, which is also capable of dealing with noisy data. [16] address low-resource challenges by developing an event detection system with minimal supervision. However, these supervised-based methods are too dependent on data. Other methods of enhancing data utilization and extrapolation are based on *transfer learning* and *pre-training*. [10] take advantage of the knowledge from data-rich classes to boost performance of data-poor ones. [41] propose a transferable architecture to transfer the knowledge from the existing event types to unseen ones. [42] use a pretraining fine-tuning paradigm to learn meaningful, interpretable prototypes for RE. [8] present a jointly extraction and generation framework to promote EE based on the pretrained language model. Though these methods are suitable for low-resource scenarios, they mostly ignore implicit correlation knowledge among classes and lack reasoning ability to some extent.

Correlation in Knowledge Extraction. In view of the shortcomings of current methods for low-resource KE, we utilize correlation knowledge to address this issue. There are some researches which try to import correlation knowledge to KE. [35] present a joint inference framework for temporal and causal relation among events by using constrained conditional models. [43] build an abstract causality network and embed it into a continuous vector space, to represent and leverage abstract event causality. [36] determine event causality by combining discourse relation predictions and distributional similarity methods in a global inference procedure. [42] propose to learn prototypes for each relation from contextual information to best explore the intrinsic semantics of relations. However, these researches utilize correlation knowledge mainly based on co-occurrence, which are dependent on sufficient data. Besides, there are also some approaches to utilizing correlation knowledge based on schema and logical rules. [44] build connections between seen and unseen relations via logic-guided semantic representation learning. [45] propose a new event graph schema to establish connections among events. [19] propose to resolve low-resource event detection problems with event ontology and first order logic, *w.r.t.* the hierarchical, temporal and causal correlation among events. [46] implement neural symbolic reasoning so as to utilize correlation knowledge between classes in KE. Although these schema-guided methods can address low-resource KE to a certain extent, the schema design is complicated and require expertise, so that limited in some domains. On the whole, current studies for utilizing correlation knowledge in KE are mostly data-dependent or schema-guided, making it difficult to utilize correlation knowledge with both effectiveness and scalability.

5. Conclusion and future work

This paper proposes a novel framework called *Knowledge-aware Hyperspherical Prototype Network (K-HPN)* to implement pairwise prototype learning on the hypersphere for low-resource knowledge extraction. In K-HPN, we represent each class as a hyperspherical prototype, and split the hyperspherical prototype network into source and target hemispheres, making it possible to utilize correlation knowledge between the source–target prototype pair. The key insight is that class-centric prototype learning should not always be data-driven, and pairwise prototype learning with correlation knowledge helps to reduce model dependence on data, especially in low-resource scenarios, as high-resource prototypes can transfer correlation knowledge to low-source ones. We have completed two tasks including relation extraction with relation entailment recognition as well as event extraction with event causality recognition, and experimental results show that K-HPN excels traditional methods for low-resource knowledge extraction.

In the future, we intend to extend our work in several aspects. First, we would consider more correlation knowledge in low-resource knowledge extraction tasks, not limited to relation entailment and event causality. Second, we would profoundly explore how low-resource knowledge extraction promotes to utilize correlation knowledge. Third, we would develop more methods based on reasoning for low-resource knowledge extraction.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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